



TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING

Flood Susceptibility Mapping of the Narayani River Basin Using Remote Sensing, GIS, and Machine Learning

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INTRODUCTION

- ❑ Floods are among the most destructive natural hazards worldwide, causing nearly 44% of all recorded disasters globally.
- ❑ Nepal's rugged topography, monsoon climate, and 6,000+ rivers make it highly flood-prone, especially June to September
- ❑ The Narayani (Gandaki) River Basin is socio-economically vital supporting agriculture, hydropower, and transport.
- ❑ Climate change, urbanization, and land-use change are intensifying flood risk across major river basins.
- ❑ Machine learning models (RF, ANN, XGBoost) can capture complex, nonlinear flood relationships better than traditional statistics.
- ❑ Remote Sensing and GIS enable large-scale, accurate analysis of flood-conditioning factors over wide areas.
- ❑ This study develops ML-based flood susceptibility maps for the Narayani River Basin to support disaster risk reduction.





Machine Learning Models

☐ Random Forest(RF)

It is an ensemble learning method that combines multiple decision trees. It is robust against overfitting, handles large datasets well, and can determine the importance of each conditioning factor.

☐ Artificial Neural Network(ANN)

It mimics the learning process of the human brain and is capable of modeling highly complex and nonlinear relationships between flood events and environmental variables.

☐ XGBoost

It is an advanced boosting algorithm that builds trees sequentially to improve prediction accuracy. It is computationally efficient and often achieves superior performance in classification problems.



OBJECTIVES

- ☐ To prepare flood conditioning factor layers.
- ☐ Create flood inventory map.
- ☐ To develop RF, ANN and XGBoost models.
- ☐ To validate and compare model performance.
- ☐ To produce flood susceptibility maps for the study area classified into five susceptibility zones (very low, low, moderate, high, very high).



STUDY AREA

The Narayani River Basin is one of Nepal's largest river systems and is highly vulnerable to monsoon-induced flooding.

- Central Nepal
- Elevation: <100 m to >8000 m
- Strong physiographic and climatic variation



SOFTWARE USED

☐ **QGIS / ArcGIS**

Spatial analysis & mapping

☐ **Google Earth Engine**

Satellite image processing

☐ **Python**

Data analysis, machine learning modeling

☐ **XGBoost / RF / ANN**

Flood prediction models

☐ **Microsoft Excel**

Data organization, preprocessing



METHODOLOGY

☐ **Data Collection**

Collected DEM, rainfall, Sentinel-2 imagery, soil data, and historical flood inventory data.

☐ **Flood Inventory Mapping**

Prepared a flood inventory map using historical flood locations.

☐ **Generate Flood Conditioning Factors**

☐ **Multicollinearity Check**

Removed highly correlated factors and retained independent variables.

☐ **Train ML Models**

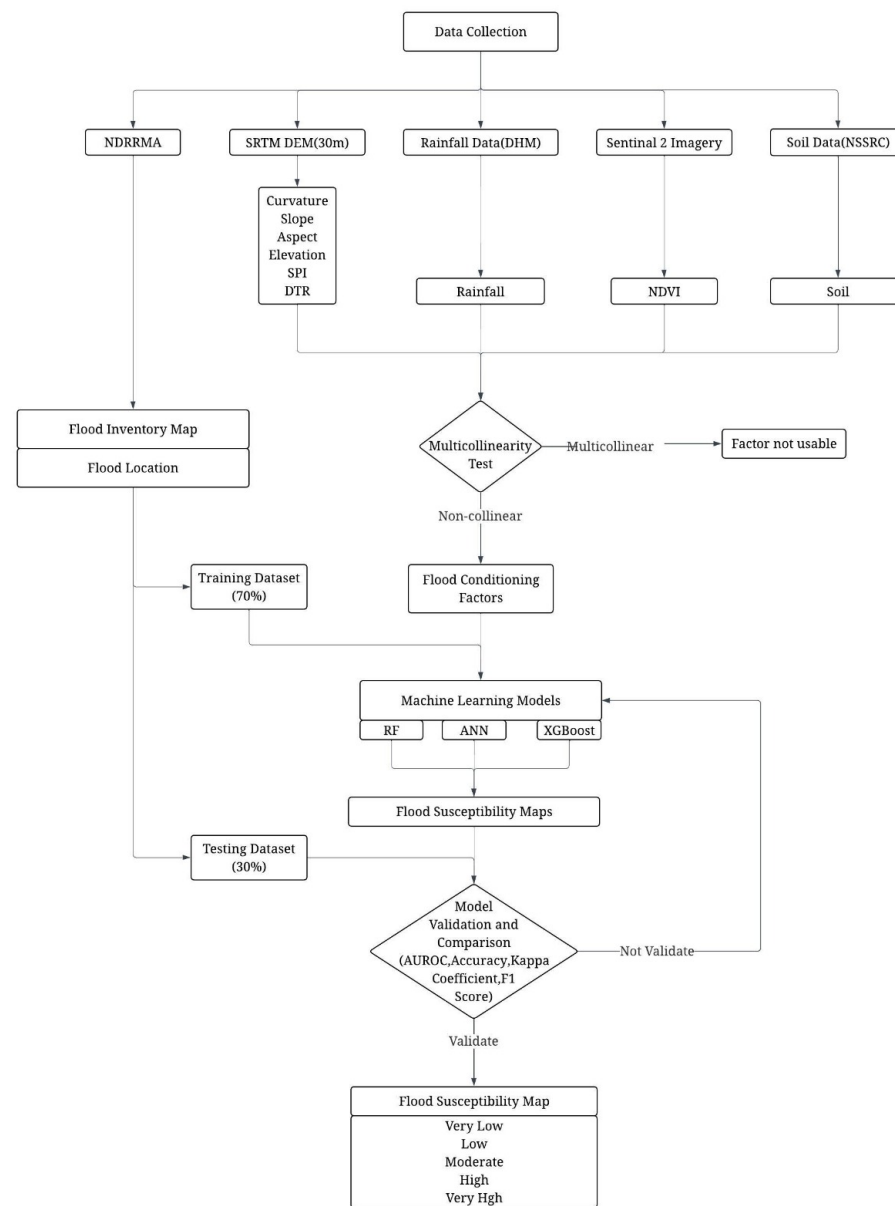
Trained RF, ANN, and XGBoost models using 70% training data and generated susceptibility maps.

☐ **Validation and Final Mapping**

Evaluated models using AUROC, Accuracy, Kappa, and F1-Score, then produced the final flood susceptibility map with five classes: Very Low, Low, Moderate, High, and Very High.



METHODOLOGY





WORK SCHEDULE AND BUDGET EST.

- ☐ The proposed research project will be executed over an 11 month timeline spanning from June to April
- ☐ three month data collection period extending from July to September.
- ☐ Data processing will then take place during October and November, leading directly into the analysis and validation phase from December to January
- ☐ The budget has been allocated across different activities necessary for the successful completion of the project.
- ☐ Transportation costs account for the largest share of the budget, with allocated for field visits, site observations, and data verification activities. Model validation has been assigned some amount to support the assessment and verification of result.



EXPECTED OUTCOMES

- ☐ Preparation of a Flood Susceptibility Map (FSM) of the Narayani River Basin.
- ☐ Determination of the most significant flood conditioning factors .
- ☐ Development and comparison of machine learning models for flood susceptibility assessment.
- ☐ Identification and classification of flood-prone areas into Very Low, Low, Moderate, High, and Very High susceptibility zones.



THANK YOU

Q & A